Coding & Solution

Date	15 March 2026
Team ID	XXXXXX
Project Name	Credit Card Fraud Detection using Machine Learning
Maximum Marks	5 Marks

Coding & Solution

This project implements Credit Card Fraud Detection using Machine Learning. It includes data preprocessing, exploratory data analysis, feature engineering, model training, evaluation and fraud prediction using Python and Scikit-learn.

Instructions

- Implemented using Python following modular programming.
- Includes preprocessing, model training and prediction modules.
- Repository contains setup instructions and source code.
- Solution follows clean coding standards and documentation.

Solution Summary

Field	Details
Repository	https://github.com/yourusername/CreditCardFraudDetection
Language	Python 3.x
Frameworks	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Features	Preprocessing, EDA, Logistic Regression, Decision Tree, Random Forest, Evaluation, Prediction
Pending	Real-time detection, Flask deployment
Run	Install requirements and run main.py

Code Quality Checklist

S.No	Criteria	Status
1	Code is modular and organized	Yes
2	Meaningful variable/function names	Yes
3	Comments/documentation included	Yes
4	Error handling implemented	Yes
5	Application runs successfully	Yes

6	GitHub version control used	Yes

Additional Notes / Comments

The project is modular, reusable, and well documented. Machine learning models are evaluated using Accuracy, Precision, Recall, F1-score and ROC-AUC. The solution can be extended for real-time fraud detection using Flask or Streamlit.